

Growatt Inverter Modbus RTU Protocol

V1.20

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Instruction: Register range for various types of inverter

TL-X (MIN Type): 03 register range: 0~124, 3000~3124; 04 register range: 3000~3124, 3125~3249

TL3-X (MAX、MID、MAC Type): 03 register range: 0~124, 125~249; 04 register range: 0~124, 125~249

Storage (MIX Type): 03 register range: 0~124, 1000~1124; 04 register range: 0~124, 1000~1124

Storage (SPA Type): 03 register range: 0~124, 1000~1124; 04 register range: 1000~1124, 2000~2124

Storage (SPH Type): 03 register range: 0~124, 1000~1124; 04 register range: 0~124, 1000~1124

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1 Data format

Address	Function	Data	CRC check
8 bits	8 bits	N×8bits	16bits

Valid slave device addresses are in the range of 0 – 254 decimal.

The individual slave devices are assigned addresses in the range of 1 – 254.

0 is the broadcast address

It is 16bits (two bytes) unsigned integer for each holding and input register;

2 Command Format

Function 3 Read holding register

QUERY	
Field Name	Example (Hex)
Slave Address	11
Function	03
Starting Address Hi	00
Starting Address Lo	6B
No. of Points Hi	00
No. of Points Lo	03
Error Check (LRC or CRC)	—

RESPONSE	
Field Name	Example (Hex)
Slave Address	11
Function	03
Byte Count	06
Data Hi (Register 40108)	02
Data Lo (Register 40108)	2B
Data Hi (Register 40109)	00
Data Lo (Register 40109)	00
Data Hi (Register 40110)	00
Data Lo (Register 40110)	64
Error Check (LRC or CRC)	—

Response Error:

11 0x80|0x03 ErrornumCRC(Errornum as a byte)

Function 4 Read input register

QUERY	
Field Name	Example (Hex)
Slave Address	11
Function	04
Starting Address Hi	00
Starting Address Lo	08
No. of Points Hi	00
No. of Points Lo	01
Error Check (LRC or CRC)	—

RESPONSE	
Field Name	Example (Hex)
Slave Address	11
Function	04
Byte Count	02
Data Hi (Register 30009)	00
Data Lo (Register 30009)	0A
Error Check (LRC or CRC)	—

Response Error:

11 0x80|0x04 ErrornumCRC (Errornum as a byte)

Function 6 Preset single register

QUERY	
Field Name	Example (Hex)
Slave Address	11
Function	06
Register Address Hi	00
Register Address Lo	01
Preset Data Hi	00
Preset Data Lo	03
Error Check (LRC or CRC)	—

RESPONSE	
Field Name	Example (Hex)
Slave Address	11
Function	06
Register Address Hi	00
Register Address Lo	01
Preset Data Hi	00
Preset Data Lo	03
Error Check (LRC or CRC)	—

Response Error:

11 0x80|0x06 ErrornumCRC (Errornum as a byte)

Function 16 Preset multiple register

QUERY	
Field Name	Example (Hex)
Slave Address	11
Function	10
Starting Address Hi	00
Starting Address Lo	01
No. of Registers Hi	00
No. of Registers Lo	02
Byte Count	04
Data Hi	00
Data Lo	0A
Data Hi	01
Data Lo	02
Error Check (LRC or CRC)	—

RESPONSE	
Field Name	Example (Hex)
Slave Address	11
Function	10
Starting Address Hi	00
Starting Address Lo	01
No. of Registers Hi	00
No. of Registers Lo	02
Error Check (LRC or CRC)	—

Response Error:

11 0x80|0x10 ErrornumCRC (Errornum as a byte)

3 Device Message Transmission Mode / Framing

RTU Mode

When controllers are setup to communicate on a Modbus network using RTU (Remote Terminal Unit) mode, each 8-bit byte in a message contains two 4-bit hexadecimal characters. Each message must be transmitted in a continuous stream.

The format for each byte in RTU mode is:

- Coding System: 8-bit binary, hexadecimal 0–9, A–F
- Two hexadecimal characters contained in each 8-bit field of the message

Bits per Byte:

- 1 start bit
- 8 data bits, least significant bit sent first
- None parity
- 1 stop bit
- Error Check Field: Cyclical Redundancy Check (CRC)

The baud rate of the transmission is:

- Default Baud Rate: 9600 bps
- Can be set through hold register 22

Minimum CMD period (RS485 Time out): 850ms.

Wait for minimum 850ms to send a new CMD after last CMD. Suggestion is 1s;

Maximum Data Length Define:

- Maximum read data length is **125 words** in read command;
- Maximum update data length is 125 words in preset command;

Note:

Except the CEIO-21 and VDE-AR-N 4105 power management registers, you should refer the manufactory's suggestion when writing other registers;

	ware	Bit0: GFCIBreak; Bit1:SPSDamage Bit8:EepromReadWarning Bit9:EEWriteWarning					
234	wCheckHardware2						reserved
235	ubNToGNDDetect	Dis/enable N to GND detect function	W	1:enable 0:disable		1	
236	NonStdVacEnable	Enable/Disable Nonstandard Grid voltage range	W	0-2		0	0:Disable; 1:Enable Voltgrade1 2:Enable Voltgrade2
237	uwEnableSpecSet	Disablse/enable appointed spec setting	W	1:enable 0:disable	Binary	0x0000	Bit 0: Hungary
238	Fast MPPT enable	About Fast mppt		0,1,2		0	Reserved
239	/	/	/	/		/	Reserved
240	Check Step		W				
241	INV-Lng	Inverter Longitude	W				Longitude
242	INV-Lat	Inverter Latitude	W				Latitude
.....							Reserved
249							Reserved

Six group for Storage Power

Register NO.	Variable Name	Description	Write or not	Value	Unit	Initial value	Note
1000.	Float charge current limit	When charge current battery need is lower than this value, enter into float charge	W		0.1A	600	CC current
1001.	PF CMD memory state	Set the following 19-22 CMD will be memory or not(1/0), if not, these settings are the initial value.	W	0or1,		0	Means these settings will be acting or not when next power on(02 repeat)
1002.	VbatStartForDischarge	LV Vbat	R/W		0.1V		Lead-acid battery LV voltage
1003.	VbatlowWarnClr	LoadPercent(only lead-Acid):	W		0.1V		Clear battery low voltage error voltage point

		45.5V <20% 48.0V 20%~50% 49.0V >50					
1004.	Vbatstopfor rdischarge	Should stop discharge when lower than this voltage(only lead-Acid): 46.0V <20% 44.8V 20%~50% 44.2V >50%	W		0.01V		
1005.	Vbat stop for charge	Should stop charge when higher than this voltage	W		0.01V	5800	
1006.	Vbat start for discharge	Should not discharge when lower than this voltage	W		0.01V	4800	
1007.	Vbat constant charge	can charge when lower than this voltage	W		0.01V	5800	CV voltage (acid)
1008.	EESysInfo.S ysSetEn	Bit0: Resved; Bit1: Resved; Bit2: Resved; Bit3: Resved; Bit4: Resved; Bit5: bDischargeEn; Bit6: ForceDischrEn; Bit7: ChargeEn; Bit8: bForceChrEn; Bit9: bBackUpEn; Bit10: bInvLimitLoadE; Bit11: bSpLimitLoadEn; Bit12: bACChargeEn; Bit13: bPVLoadLimitEn; Bit14,15:Unused;	W				System Enable
1009.	Battemp lower limit d	Battery temperature lower limit for discharge	W	0-200:0-2 0℃ 1000-140	0.1℃	1170	

				0: -40-0℃			
1010.	Bat temp upper limit	Battery temperature upper limit for discharge	W	200-1000	0.1℃	420	
1011.	Bat temp lower limit	Battery temperature lower limit for charge	W	0-200:0-20℃ 1000-1400: -40-0℃	0.1℃	30	Lower temperature limit
1012.	Bat temp upper limit	Battery temperature upper limit for charge	W	200-1000	0.1℃	370	Upper temperature limit
1013.	uwUnderFreDischargeDelyTime	Under Fre Delay Time	s	0-20	50ms		Under Fre Delay Time
1014.	BatMdlSerialNum	Battery serial number	W	00:00			SPH4-11K used
1015.	BatMdlParallelNum	Battery parallel section	W	00:00			SPH4-11K used
1016.	/	/	/	/	/	/	Reserve
1017.	/	/	/	/	/	/	Reserve
1018.	/	/	/	/	/	/	Reserve
1019.	/	/	/	/	/	/	Reserve
1020.	/	/	/	/	/	/	Reserve
1021.	/	/	/	/	/	/	Reserve
1022.	/	/	/	/	/	/	Reserve
1023.	/	/	/	/	/	/	Reserve
1024.	/	/	/	/	/	/	Reserve
1025.	/	/	/	/	/	/	Reserve

1026.	/	/	/	/	/	/	Reserve
1027.	/	/	/	/	/	/	Reserve
1028.	/	/	/	/	/	/	Reserve
1029.	/	/	/	/	/	/	Reserve
1030.	/	/	/	/	/	/	Reserve
1031.	/	/	/	/	/	/	Reserve
1032.	/	/	/	/	/	/	Reserve
1033.	/	/	/	/	/	/	Reserve
1034.	/	/	/	/	/	/	Reserve
1035.	/	/	/	/	/	/	Reserve
1036.	/	/	/	/	/	/	Reserve
1037.	bCTMode	Use the CTMode to Choose RFCT \ Cable CT\METER	W	2:METER 1:cWirele ssCT 0:cWiredC T		0	
1038.	CTAdjust	CTAdjust enable	W	0:disable 1:enable		0	
1039.	/	/	/	/	/	/	Reserve
1040.	/	/	/	/	/	/	Reserve
1041.	/	/	/	/	/	/	Reserve
1042.	/	/	/	/	/	/	Reserve
1043.	/	/	/	/	/	/	Reserve

1044.	Priority	ForceChrEn/ForceDischrEn Load first/bat first /grid first	R	0:Load(default)/1.Battery/2.Grid			bForceChrEn/disbForceDischrEn/dis
1045.	/	/	/	/	/	/	Reserve
1046.	/	/	/	/	/	/	Reserve
1047.	AgingTestStep Cmd	Command for aging test		0: default 1: charge 2: discharge			Cmd for aging test
1048.	BatteryType	Battery type choose of buck-boost input		0:Lithium 1:Lead-acid 2:other		0	Battery type
1049.	/	/	/	/	/		Reserve
1050.	/	/	/	/	/	/	Reserve
1051.	/	/	/	/	/		Reserve
1052.	/	/	/	/	/		Reserve
1053.	/	/	/	/	/		Reserve
1054.	/	/	/	/	/	/	Reserve
1060.	BuckUpsFunction	Ups function enable or disable		0:disable 1:enable			
1061.	BuckUPSVoltage	UPS output voltage		0:230 1:208 2:240		230V	
1062.	UPSFreqSet	UPS output frequency		0:50Hz 1:60Hz		50Hz	
...	/	/	/	/	/	/	reverse
Priority set							

1070.	GridFirstDischargePowerRate	Discharge Power Rate when Grid First	W	0-100	1%	Discharge Power Rate when Grid First	
1071.	GridFirstStopSOC	Stop Discharge soc when Grid First	W	0-100	1%	Stop Discharge soc when Grid First	
1072... 1079	/	/	/	/	/	/	reverse
1080.	Grid First Start Time 1	High eight bit: hour Low eight bit: minute		0-23 0-59			
1081.	Grid First Stop Time 1	High eight bit: hour Low eight bit: minute		0-23 0-59			
1082.	Grid First Stop Switch 1	Enable :1 Disable:0		0 or 1		Grid First enable	
1083.	Grid First Start Time 2	High eight bit: hour Low eight bit: minute		0-23 0-59			
1084.	Grid First Stop Time 2	High eight bit: hour Low eight bit: minute		0-23 0-59			
1085.	Grid First Stop Switch 2	ForceDischarge.bSwitch&LCD_SET_FORCE_TRUE_2)=LCD_SET_FORCE_TRUE_2		0 or 1		Grid First enable	ForceDischarge; LCD_SET_FORCE_TRUE_2
1086.	Grid First Start Time 3	High eight bit: hour Low eight bit: minute		0-23 0-59			
1087.	Grid First Stop Time 3	High eight bit: hour Low eight bit: minute		0-23 0-59			
1088.	Grid First Stop Switch 3	Enable :1 Disable:0		0 or 1		Grid First enable	
1089.	/	/	/	/	/	/	reserve
1090.	BatFirstPowerRate	Charge Power Rate when Bat First	W	0-100	1%	Charge Power Rate when Bat First	
1091.	wBatFirst stop SOC	Stop Charge soc when Bat First	W	0-100	1%	Stop Charge soc when Bat First	

1092.	AC charge Switch	When Bat First Enable:1 Disable:0		Enable:1 Disable:0		AC Charge Enable	
1093... 1099							
1100.	Bat First Start Time 1	High eight bit: hour Low eight bit: minute		0-23 0-59			
1101.	Bat First Stop Time 1	High eight bit: hour Low eight bit: minute		0-23 0-59			
1102.	BatFirst on/off Switch 1	Enable :1 Disable:0		0 or 1		Bat First Enable1	
1103.	Bat First Start Time 2	High eight bit: hour Low eight bit: minute		0-23 0-59			
1104.	Bat First Stop Time 2	High eight bit: hour Low eight bit: minute		0-23 0-59			
1105.	BatFirston/off Switch 2	Enable :1 Disable:0		0 or 1		Bat First Enable2	
1106.	Bat First Start Time 3	High eight bit: hour Low eight bit: minute		0-23 0-59			
1107.	Bat First Stop Time 3	High eight bit: hour Low eight bit: minute		0-23 0-59			
1108.	BatFirston/off Switch 3	Enable :1 Disable:0		0 or 1		Bat First Enable3	
1109.	/	/	/	/	/	/	reserve
1110.	Load First Start Time 1	High eight bit: hour Low eight bit: minute		0-23 0-59			SPA/ reserve
1111.	Load First Stop Time 1	High eight bit: hour Low eight bit: minute		0-23 0-59			SPA/ reserve
1112.	Load First Switch 1	Enable :1 Disable:0		0 or 1		Load First Enable	SPA/ reserve
1113.	Load First Start Time2	High eight bit: hour Low eight bit: minute		0-23 0-59			SPA/ reserve
1114.	Load First Stop Time 2	High eight bit: hour Low eight bit: minute		0-23 0-59			SPA/ reserve
1115.	Load First Switch 2	Enable :1 Disable:0		0 or 1		Load First Enable	SPA/ reserve

1116.	Load First Start Time 3	High eight bit: hour Low eight bit: minute		0-23 0-59			SPA/ reserve
1117.	Load First Stop Time 3	High eight bit: hour Low eight bit: minute		0-23 0-59			SPA/ reserve
1118.	Load First Switch 3	Enable :1 Disable:0		0 or 1		Load First Enable	SPA/ reserve
1119.	/	/	/	/	/	/	reserve
1120.	BackUpEn	BackUp Enable					MIX US
1121.	SGIPEn	SGIP Enable					MIX US
.....	1122~1124	/	/	/	/	/	reserve
Use for TL-X and TL-XH							
3000	ExportLimitFa iledPowerRat e	The power rate when exportLimit failed	R/W		0.1%		The power rate when exportLimit failed
3001	New Serial NO	Serial number 1-2	R/W		ASCII		The new model uses the following registers to record the serial number; The representation is the same as the original: one register holds two characters and the new serial number is 30 characters.
3002	New Serial NO	Serial number 3-4	R/W		ASCII		
3003	New Serial NO	Serial number 5-6	R/W		ASCII		
3004	New Serial NO	Serial number 7-8	R/W		ASCII		
3005	New Serial NO	Serial number 9-10	R/W		ASCII		
3006	New Serial NO	Serial number 11-12	R/W		ASCII		
3007	New Serial NO	Serial number 13-14	R/W		ASCII		
3008	New Serial NO	Serial number 15-16	R/W		ASCII		
3009	New Serial NO	Serial number 17-18	R/W		ASCII		
3010	New Serial NO	Serial number 19-20	R/W		ASCII		
3011	New Serial NO	Serial number 21-22	R/W		ASCII		

3012	New Serial NO	Serial number 23-24	R/W		ASCII		
3013	New Serial NO	Serial number 25-26	R/W		ASCII		
3014	New Serial NO	Serial number 27-28	R/W		ASCII		
3015	New Serial NO	Serial number 29-30	R/W		ASCII		
3016	DryContactFunctionEn	DryContact function enable	R/W	0:Disable 1: Enable			DryContact function enable
3017	DryContactOnRate	The power rate of drycontact turn on	R/W	0~1000	0.1%		The power rate of drycontact turn on
3018	Reserved						
3019	DryContactOffRate	DryContactOffRate	Dry contact closure power	R/W	0~100 0	0.1%	Dry contact closure power percentage
3020	BoxCtrlInvOrder	BoxCtrlInvOrder	Off-net box control instruction	R/W			
3021	ExterCommOffGridEn	External communication setting manual off-network enable	R/W				0x00: Disable; (default) 0x01: Enable;
3022	Reserved						
3023	Reserved						
3024	Float charge current limit	When charge current battery need is lower than this value, enter into float charge	R/W		0.1A	600	CC current
3025	VbatWarning	"Battery-low" warning setup voltage	R/W		0.1V	4800	Lead acid battery LV voltage
3026	VbatlowWarnClr	"Battery-low" warning clear voltage	R/W		0.1V		Clear battery low voltage error voltage point LoadPercent(only lead-Acid):

							45.5V(Load < 20%); 48.0V(20%≤Load ≤50%); 49.0V(Load > 50%);
3027	Vbatstopfordi scharge	Battery cut off voltage	R/W		0.1V		Should stop discharge when lower than this voltage(only lead-Acid): 46.0V(Load < 20%); 44.8V(20%≤Load ≤50%); 44.2V(Load > 50%);
3028	Vbat stop for charge	Battery over charge voltage	R/W		0.01V	5800	Should stop charge when higher than this voltage
3029	Vbat start for discharge	Battery start discharge voltage	R/W		0.01V	4800	Should not discharge when lower than this voltage
3030	Vbat constant charge	Battery constant charge voltage	R/W		0.01V	5800	CV voltage (acid) can charge when lower than this voltage
3031	Battemp lower limit d	Battery temperature lower limit for discharge	R/W		0.1℃	1170	0-200:0-20℃ 1000-1400: -40-0℃
3032	Bat temp upper limit d	Battery temperature upper limit for discharge	R/W		0.1℃	420	
3033	Bat temp lower limit c	Battery temperature lower limit for charge	R/W		0.1℃	30	Battery temperature lower limit 0-200:0-20℃ 1000-1400: -40-0℃

3034	Bat temp upper limit c	Battery temperature upper limit for charge	R/W		0.1℃	370	Battery temperature upper limit
3035	uwUnderFreDischargeDelayTime	Under Fre Delay Time	R/W		50ms		Under Fre Delay Time
3036	GridFirstDischargePowerRate	Discharge Power Rate when Grid First				1-255	
3037	GridFirstStopSOC	Stop Discharge soc when Grid First				1-100	
3038	Time 1(xh)	Period 1: [Start Time ~ End Time], [Charge/Discharge], [Disable/Enable] 3038 enable, charge and discharge, start time, end time 3039	R/W				Bit0~7: minutes; Bit8~12: hour; Bit13~14, 0: load priority; 1: battery priority; 2: Grid priority; Bit15, 0: prohibited; 1: enabled;
3039			R/W				Bit0~7: minutes; Bit8~12: hour; Bit13~15: reserved
3040	Time 2(xh)	Time period 2: [start time ~ end time], [charge / discharge], [disable / enable] 3040 enable, charge and discharge, start time, 3041 end time	R/W				Bit0~7: minutes; Bit8~12: hour; Bit13~14, 0: load priority; 1: battery priority; 2: Grid priority; Bit15, 0: prohibited; 1: enabled;
3041			R/W				Bit0~7: minutes; Bit8~12: hour; Bit13~15: reserved
3042	Time 3(xh)	With Time1	R/W				With Time1
3043			R/W				With Time1
3044	Time 4(xh)	With Time1	R/W				With Time1
3045			R/W				With Time1
3046	Grid First Stop Switch 3	Grid first time-3 enable					Enable :1 Disable:0

3047	BatFirstPowerRate	Charge Power Rate when Bat First				1-100	
3048	wBatFirst stop SOC	Stop Charge soc when Bat First				1-100	
3049	AcChargeEnable	AcChargeEnable					Enable :1 Disable:0
3050	Time 5(xh)	同 Time1	R/W				With Time1
3051			R/W				With Time1
3052	Time 6(xh)	同 Time1	R/W				With Time1
3053			R/W				With Time1
3054	Time 7(xh)	同 Time1	R/W				With Time1
3055			R/W				With Time1
3056	Time 8(xh)	同 Time1	R/W				With Time1
3057			R/W				With Time1
3058	Time 9(xh)	同 Time1	R/W				With Time1
3059			R/W				With Time1
3060~ 3069	Reserved						
3070	BatteryType	Battery type choose of buck-boost input	R/W				Battery type 0:Lithium 1:Lead-acid 2:other
3071	BatMdlSeria/ ParalNum	BatMdlSeria/ParalNum	R/W				BatMdlSeria/Paral Num; SPH4-11K used The upper 8 bits indicate the number of series segments; The lower 8 bits

							indicate the number of parallel sections;
3072	Reserved						
3073	Reserved						
3074	Reserved						
3075	Reserved						
3076	Reserved						
3077	Reserved						
3078	Reserved						
3079	UpsFunEn	Ups function enable or disable	R/W			0	0:disable 1:enable
3080	UPSVoltSet	UPS output voltage	R/W			0	0:230V 1:208V 2:240V
3081	UPSFreqSet	UPS output frequency	R/W			0	0:50Hz 1:60Hz
3082	bLoadFirstStopSocSet	StopSoc When LoadFirst	R/W			13-100	ratio
3083	Reserved						
3084	Reserved						
3085	Com Address	Communication addr	R/W			1	1 : Communication addr=1 1 ~ 254 : Communication addr=1~254
3086	BaudRate	Communication BaudRate	R/W			0	0: 9600 bps 1: 38400 bps
3087	Serial NO	Serial Number 1-2	R/W		ASCII		For battery
3088	Serial No	Serial Number 3-4	R/W		ASCII		
3089	Serial No	Serial Number 5-6	R/W		ASCII		
3090	Serial No	Serial Number 7-8	R/W		ASCII		
3091	Serial No	Serial Number 9-10	R/W		ASCII		
3092	Model H	Model H	R/W				
3093	Model L	Model L	R/W				
3094	Pdischr max H	Max Discharge Power	R		0.1W		
3095	Pdischr max L						
3096	Pchr max H	Max Charge Power	R		0.1W		
3097	Pchr max L						
3098	DTC	DTC	R				

3099	FW Code1	FW Code1	R		ASCII		
3100	FW Code2	FW Code2	R		ASCII		
3101	Processor1 FW Vision	Processor1 FW Vision	R		ASCII		
3102	Reset User Info	Reset User Info	W				
3103	Reset to factory	Reset to factory	W				
3104 ~ 3124	Reserved						

4.2 Input Reg

NO.	Variable Name	Description	Value	Unit	Note
First group					
0.	Inverter Status	Inverter run state	0:waiting, 1:normal, 3:fault		
1.	Ppv H	Input power (high)		0.1W	
2.	Ppv L	Input power (low)		0.1W	
3.	Vpv1	PV1 voltage		0.1V	
4.	PV1Curr	PV1 input current		0.1A	
5.	Ppv1 H	PV1 input power(high)		0.1W	
6.	Ppv1 L	PV1 input power(low)		0.1W	
7.	Vpv2	PV2 voltage		0.1V	
8.	PV2Curr	PV2 input current		0.1A	
9.	Ppv2 H	PV2 input power (high)		0.1W	
10.	Ppv2 L	PV2 input power (low)		0.1W	
11.	Vpv3	PV3 voltage		0.1V	
12.	PV3Curr	PV3 input current		0.1A	
13.	Ppv3 H	PV3 input power (high)		0.1W	
14.	Ppv3 L	PV3 input power (low)		0.1W	
15.	Vpv4	PV4 voltage		0.1V	
16.	PV4Curr	PV4 input current		0.1A	
17.	Ppv4 H	PV4 input power (high)		0.1W	
18.	Ppv4 L	PV4 input power (low)		0.1W	
19.	Vpv5	PV5 voltage		0.1V	
20.	PV5Curr	PV5 input current		0.1A	
21.	Ppv5H	PV5 input power(high)		0.1W	

22.	Ppv5 L	PV5 input power(low)		0.1W	
23.	Vpv6	PV6 voltage		0.1V	
24.	PV6Curr	PV6 input current		0.1A	
25.	Ppv6 H	PV6 input power (high)		0.1W	
26.	Ppv6 L	PV6 input power (low)		0.1W	
27.	Vpv7	PV7 voltage		0.1V	
28.	PV7Curr	PV7 input current		0.1A	
29.	Ppv7 H	PV7 input power (high)		0.1W	
30.	Ppv7 L	PV7 input power (low)		0.1W	
31.	Vpv8	PV8 voltage		0.1V	
32.	PV8Curr	PV8 input current		0.1A	
33.	Ppv8 H	PV8 input power (high)		0.1W	
34.	Ppv8 L	PV8 input power (low)		0.1W	
35.	Pac H	Output power (high)		0.1W	
36.	Pac L	Output power (low)		0.1W	
37.	Fac	Grid frequency		0.01Hz	
38.	Vac1	Three/single phase grid voltage		0.1V	
39.	Iac1	Three/single phase grid output current		0.1A	
40.	Pac1 H	Three/single phase grid output watt VA (high)		0.1VA	
41.	Pac1 L	Three/single phase grid output watt VA(low)		0.1VA	
42.	Vac2	Three phase grid voltage		0.1V	
43.	Iac2	Three phase grid output current		0.1A	
44.	Pac2 H	Three phase grid output power (high)		0.1VA	
45.	Pac2 L	Three phase grid output power (low)		0.1VA	
46.	Vac3	Three phase grid voltage		0.1V	
47.	Iac3	Three phase grid output current		0.1A	
48.	Pac3 H	Three phase grid output power (high)		0.1VA	
49.	Pac3 L	Three phase grid output power (low)		0.1VA	
50.	Vac_RS	Three phase grid voltage		0.1V	Line voltage
51.	Vac_ST	Three phase grid voltage		0.1V	Line voltage
52.	Vac_TR	Three phase grid voltage		0.1V	Line voltage
53.	Eactoday H	Today generate energy (high)		0.1kWH	
54.	Eac today L	Today generate energy (low)		0.1kWH	
55.	Eac total H	Total generate energy (high)		0.1kWH	
56.	Eac total L	Total generate energy (low)		0.1kWH	
57.	Time total H	Work time total (high)		0.5s	
58.	Time total L	Work time total (low)		0.5s	
59.	Epv1_today H	PV1Energy today(high)		0.1kWh	
60.	Epv1_today L	PV1Energy today (low)		0.1kWh	

61.	Epv1_total H	PV1Energy total(high)		0.1kWh	
62.	Epv1_total L	PV1Energy total (low)		0.1kWh	
63.	Epv2_today H	PV2Energy today(high)		0.1kWh	
64.	Epv2_today L	PV2Energy today (low)		0.1kWh	
65.	Epv2_total H	PV2Energy total(high)		0.1kWh	
66.	Epv2_total L	PV2Energy total (low)		0.1kWh	
67.	Epv3_today H	PV3 Energy today(high)		0.1kWh	
68.	Epv3_today L	PV3 Energy today (low)		0.1kWh	
69.	Epv3_total H	PV3 Energy total(high)		0.1kWh	
70.	Epv3_total L	PV3 Energy total (low)		0.1kWh	
71.	Epv4_today H	PV4Energy today(high)		0.1kWh	
72.	Epv4_today L	PV4Energy today (low)		0.1kWh	
73.	Epv4_total H	PV4Energy total(high)		0.1kWh	
74.	Epv4_total L	PV4Energy total (low)		0.1kWh	
75.	Epv5_today H	PV5Energy today(high)		0.1kWh	
76.	Epv5_today L	PV5Energy today (low)		0.1kWh	
77.	Epv5_total H	PV5Energy total(high)		0.1kWh	
78.	Epv5_total L	PV5Energy total (low)		0.1kWh	
79.	Epv6_today H	PV6Energy today(high)		0.1kWh	
80.	Epv6_today L	PV6Energy today (low)		0.1kWh	
81.	Epv6_total H	PV6Energy total(high)		0.1kWh	
82.	Epv6_total L	PV6Energy total (low)		0.1kWh	
83.	Epv7_today H	PV7Energy today(high)		0.1kWh	
84.	Epv7_today L	PV7Energy today (low)		0.1kWh	
85.	Epv7_total H	PV7 Energy total(high)		0.1kWh	
86.	Epv7_total L	PV7Energy total (low)		0.1kWh	
87.	Epv8_today H	PV8Energy today(high)		0.1kWh	
88.	Epv8_today L	PV8Energy today (low)		0.1kWh	
89.	Epv8_total H	PV8Energy total(high)		0.1kWh	
90.	Epv8_total L	PV8Energy total (low)		0.1kWh	
91.	Epv_total H	PV Energy total(high)		0.1kWh	
92.	Epv_total L	PV Energy total (low)		0.1kWh	
93.	Temp1	Inverter temperature		0.1C	
94.	Temp2	The inside IPM in inverter Temperature		0.1C	
95.	Temp3	Boost temperature		0.1C	
96.	Temp4				reserved
97.	uwBatVolt_DSP	BatVolt_DSP		0.1V	BatVolt(DSP)
98.	P Bus Voltage	P Bus inside Voltage		0.1V	
99.	N Bus Voltage	N Bus inside Voltage		0.1V	

100.	IPF	Inverter output PF now	0-20000		
101.	RealOPPercent	Real Output power Percent		1%	
102.	OPFullwatt H	Output Maxpower Limited high			
103.	OPFullwatt L	Output Maxpower Limited low		0.1W	
104.	DeratingMode	DeratingMode	0:no derate; 1:PV; 2:*; 3:Vac; 4:Fac; 5:Tboost; 6:Tinv; 7:Control; 8:*; 9:*OverBack ByTime;		
105.	Fault code	Inverter fault code	&*1		MAX
106.	Fault Bitcode H	Inverter fault code high	&*8		MAX
107.	Fault Bitcode L	Inverter fault code low			
108.	RemoteCtrlEn	/	0.Load First	/	StoragePower (SPA)
109.	RemoteCtrlPower	/	1.BatFirst 2.Grid	/	StoragePower (SPA)
110.	Warning bit H	Warning bit H	&*8		
111.	Warning bit L	Warning bit L			
112.	bINVWarnCode	bINVWarnCode			MAX
	EACCharge_Today_H	ACCharge energy today		0.1kwh	Storage Power
113.	real Power Percent	real Power Percent	0-100	%	MAX
	EACCharge_Today_L	ACCharge energy today		0.1kwh	Storage Power
114.	inv start delay time	inv start delay time			MAX
	EACCharge_Total_H	ACCharge energy total		0.1kwh	Storage Power
115.	bINVAllFaultCode	bINVAllFaultCode			MAX
	EACCharge_Total_L	ACCharge energy total		0.1kwh	Storage Power
116.	AC charge Power_H	Grid power to local load		0.1kwh	Storage Power

117.	AC charge Power_L	Grid power to local load		0.1kwh	Storage Power
118.	Priority	0:Load First 1:Battery First 2:Grid First			Storage Power
119.	Battery Type	0: Lead-acid 1: Lithium battery			Storage Power
120.	AutoProofreadC MD	Aging mode 自动校准命令			Storage Power
...	reserved				reserved
124.	reserved				reserved
Second group					
125.	PID PV1+ Voltage	PID PV1PE Volt	0~1000V	0.1V	
126.	PID PV1+ Current	PID PV1PE Curr	-10~10mA	0.1mA	
127.	PID PV2+ Voltage	PID PV2PE Volt	0~1000V	0.1V	
128.	PID PV2+ Current	PID PV2PE Curr	-10~10mA	0.1mA	
129.	PID PV3+ Voltage	PID PV3PE Volt	0~1000V	0.1V	
130.	PID PV3+ Current	PID PV3PE Curr	-10~10mA	0.1mA	
131.	PID PV4+ Voltage	PID PV4PE Volt	0~1000V	0.1V	
132.	PID PV4+ Current	PID PV4PE Curr	-10~10mA	0.1mA	
133.	PID PV5+ Voltage	PID PV5PE Volt	0~1000V	0.1V	
134.	PID PV5+ Current	PID PV5PE Curr	-10~10mA	0.1mA	
135.	PID PV6+ Voltage	PID PV6PE Volt	0~1000V	0.1V	
136.	PID PV6+ Current	PID PV6PE Curr	-10~10mA	0.1mA	
137.	PID PV7+ Voltage	PID PV7PE Volt	0~1000V	0.1V	
138.	PID PV7+ Current	PID PV7PE Curr	-10~10mA	0.1mA	
139.	PID PV8+ Voltage	PID PV8PE Volt	0~1000V	0.1V	
140.	PID PV8+ Current	PID PV8PE Curr	-10~10mA	0.1mA	
141.	PID Status	Bit0~7:PID Working Status 1:Wait Status 2:Normal Status 3:Fault Status Bit8~15:Reversed	0~3		
142.	V _String1	PV String1 voltage		0.1V	
143.	Curr _String1	PV String1 current	-15~15A	0.1A	
144.	V _String2	PV String2 voltage		0.1V	
145.	Curr _String2	PV String2 current	-15~15A	0.1A	
146.	V _String3	PV String3 voltage		0.1V	
147.	Curr _String3	PV String3 current	-15~15A	0.1A	
148.	V _String4	PV String4 voltage		0.1V	

149.	Curr_String4	PV String4 current	-15~15A	0.1A	
150.	V_String5	PV String5 voltage		0.1V	
151.	Curr_String5	PV String5 current	-15~15A	0.1A	
152.	V_String6	PV String6 voltage		0.1V	
153.	Curr_String6	PV String6 current	-15~15A	0.1A	
154.	V_String7	PV String7 voltage		0.1V	
155.	Curr_String7	PV String7 current	-15~15A	0.1A	
156.	V_String8	PV String8 voltage		0.1V	
157.	Curr_String8	PV String8 current	-15A~15A	0.1A	
158.	V_String9	PV String9 voltage		0.1V	
159.	Curr_String9	PV String9 current	-15A~15A	0.1A	
160.	V_String10	PV String10 voltage		0.1V	
161.	Curr_String10	PV String10 current	-15~15A	0.1A	
162.	V_String11	PV String11 voltage		0.1V	
163.	Curr_String11	PV String11 current	-15~15A	0.1A	
164.	V_String12	PV String12 voltage		0.1V	
165.	Curr_String12	PV String12 current	-15~15A	0.1A	
166.	V_String13	PV String13 voltage		0.1V	
167.	Curr_String13	PV String13 current	-15A~15A	0.1A	
168.	V_String14	PV String14 voltage		0.1V	
169.	Curr_String14	PV String14 current	-15~15A	0.1A	
170.	V_String15	PV String15 voltage		0.1V	
171.	Curr_String15	PV String15 current	-15~15A	0.1A	
172.	V_String16	PV String16 voltage		0.1V	
173.	Curr_String16	PV String16 current	-15~15A	0.1A	
174.	StrUnmatch	Bit0~15: String1~16 unmatch			suggestive
175.	StrCurrentUnbalance	Bit0~15: String1~16 current unbalance			suggestive
176.	StrDisconnect	Bit0~15: String1~16 disconnect			suggestive
177.	PIDFaultCode	Bit0: Output over voltage Bit1: ISO fault Bit2: BUS voltage abnormal Bit3~15: reserved			
178.	String Prompt	String Prompt Bit0: String Unmatch Bit1: StrDisconnect Bit2: StrCurrentUnbalance Bit3~15: reserved			
179.	PV Warning Value	PV Warning Value			
180.	DSP075 Warning Value	DSP075 Warning Value			

181	DSP075 Fault Value	DSP075 Fault Value			
182	DSP067 Debug Data1	DSP067 Debug Data1			
183	DSP067 Debug Data2	DSP067 Debug Data2			
184	DSP067 Debug Data3	DSP067 Debug Data3			
185	DSP067 Debug Data4	DSP067 Debug Data4			
186	DSP067 Debug Data5	DSP067 Debug Data5			
187	DSP067 Debug Data6	DSP067 Debug Data6			
188	DSP067 Debug Data7	DSP067 Debug Data7			
189	DSP067 Debug Data8	DSP067 Debug Data8			
190	DSP075 Debug Data1	DSP075 Debug Data1			
191	DSP075 Debug Data2	DSP075 Debug Data2			
192	DSP075 Debug Data3	DSP075 Debug Data3			
193	DSP075 Debug Data4	DSP075 Debug Data4			
194	DSP075 Debug Data5	DSP075 Debug Data5			
195	DSP075 Debug Data6	DSP075 Debug Data6			
196	DSP075 Debug Data7	DSP075 Debug Data7			
197	DSP075 Debug Data8	DSP075 Debug Data8			
198	bUSBAgingTestOk Flag	USBAgingTestOkFlag	0-1		
199	bFlashEraseAging OkFlag	FlashEraseAgingOkFlag	0-1		
200	PVISO	PVISOValue		K Ω	
201	R_DCI	R DCI Curr		0.1mA	
202	S_DCI	S DCI Curr		0.1mA	
203	T_DCI	T DCI Curr		0.1mA	

204	PID_Bus	PIDBusVolt		0.1V	
205	GFCI	GFCI Curr		mA	
206	SVG/APF Status+SVGAPFEq ualRatio	SVG/APF Status+SVGAPFEqualRatio	High 8 bit : SVGAPFEqua lRatio Low 8 bit : SVG/APF Status 0:None 1:SVG Run 2:APF Run 3:SVG/APF Run		
207	CT_I_R	R phase load side current for SVG		0.1A	
208	CT_I_S	S phase load side current for SVG		0.1A	
209	CT_I_T	T phase load side current for SVG		0.1A	
210	CT_Q_R H	R phase load side output reactive power for SVG(High)		0.1Var	
211	CT_Q_R L	R phase load side output reactive power for SVG(low)		0.1Var	
212	CT_Q_S H	S phase load side output reactive power for SVG(High)		0.1Var	
213	CT_Q_S L	S phase load side output reactive power for SVG(low)		0.1Var	
214	CT_Q_T H	T phase load side output reactive power for SVG(High)		0.1Var	
215	CT_Q_T L	T phase load side output reactive power for SVG(low)		0.1Var	
216	CT HAR_I_R	R phase load side harmonic		0.1A	
217	CT HAR_I_S	S phase load side harmonic		0.1A	
218	CT HAR_I_T	T phase load side harmonic		0.1A	
219	COMP_Q_R H	R phase compensate reactive power for SVG(High)		0.1Var	
220	COMP_Q_R L	R phase compensate reactive power for SVG(low)		0.1Var	
221	COMP_Q_S H	S phase compensate reactive power for SVG(High)		0.1Var	
222	COMP_Q_S L	S phase compensate reactive power for SVG(low)		0.1Var	
223	COMP_Q_T H	T phase compensate reactive power for SVG(High)		0.1Var	
224	COMP_Q_T L	T phase compensate reactive power		0.1Var	

		for SVG(low)			
225	COMP HAR_I_R	R phase compensate harmonic for SVG		0.1A	
226	COMP HAR_I_S	S phase compensate harmonic for SVG		0.1A	
227	COMP HAR_I_T	T phase compensate harmonic for SVG		0.1A	
228	bRS232AgingTestOkFlag	RS232AgingTestOkFlag	0-1		
229	bFanFaultBit	Bit0: Fan 1 fault bit Bit1: Fan 2 fault bit Bit2: Fan 3 fault bit Bit3: Fan 4 fault bit Bit4-7: Reserved			
230	Sac H	Output apparent power H		0.1W	
231	Sac L	Output apparent power L		0.1W	
232	ReActPowerH	Real Output Reactive Power H	Int32	0.1W	
233	ReActPowerL	Real Output Reactive Power L			
234	ReActPowerMaxH	Nominal Output Reactive Power H		0.1var	
235	ReActPowerMaxL	Nominal Output Reactive Power L			
236	ReActPower_Total H	Reactive power generation		0.1kwh	
237	ReActPower_Total L	Reactive power generation			
...	238~249				reserved
Ninth group for Storage power					
1000.	uwSysWorkMode	System work mode	0x00:waiting module 0x01: Self-test mode, optional 0x02 : Reserved 0x03 : SysFault module 0x04: Flash module 0x05 : PVBATOnline module, 0x06 : BatOnline		The working mode displayed by the monitoring to the customer is: 0x00: waiting module 0x01: Self-test mode, 0x03: fault module 0x04: flash module 0x05 0x06 0x07 0x08: normal module

			module, 0x07 : PVOOfflineMod e module, 0x08 : BatOfflineMo de module,		
1001.	Systemfault word0	System fault word0			Please refer to thefault description of Hybrid
1002.	Systemfault word1	System fault word1			
1003.	Systemfault word2	System fault word2			
1004.	Systemfault word3	System fault word3			
1005.	Systemfault word4	System fault word4			
1006.	Systemfault word5	System fault word5			
1007.	Systemfault word6	System fault word6			
1008.	Systemfault word7	System fault word7			
1009.	Pdischarge1 H	Discharge power(high)		0.1W	
1010.	Pdischarge1 L	Discharge power (low)		0.1W	
1011.	Pcharge1 H	Charge power(high)		0.1W	
1012.	Pcharge1 L	Charge power (low)		0.1W	
1013.	Vbat	Battery voltage		0.1V	
1014.	SOC	State of charge Capacity	0-100	1%	lith/leadacid
1015.	Pactouser R H	AC power to user H		0.1w	
1016.	Pactouser R L	AC power to user L		0.1w	
1017.	Pactouser S H	Pactouser S H		0.1w	
1018.	Pactouser S L	Pactouser S L		0.1w	
1019.	Pactouser T H	Pactouser T H		0.1w	
1020.	Pactouser T L	Pactouser T H		0.1w	
1021.	PactouserTotal H	AC power to user total H		0.1w	
1022.	PactouserTotal L	AC power to user total L		0.1w	
1023.	Pac to grid R H	AC power to grid H		0.1w	Ac output
1024.	Pac to grid R L	AC power to grid L		0.1w	
1025.	Pactogrid S H			0.1w	
1026.	Pactogrid S L			0.1w	
1027.	Pactogrid T H			0.1w	
1028.	Pactogrid T L			0.1w	
1029.	Pactogrid total H	AC power to grid total H		0.1w	
1030.	Pactogrid total L	AC power to grid total L		0.1w	
1031.	PLocalLoad R H	INV power to local load H		0.1w	

1032.	PLocalLoad R L	INV power to local load L		0.1w	
1033.	PLocalLoad S H			0.1w	
1034.	PLocalLoad S L			0.1w	
1035.	PLocalLoadT H			0.1w	
1036.	PLocalLoadT L			0.1w	
1037.	PLocalLoad total H	INV power to local load total H		0.1w	
1038.	PLocalLoad total L	INV power to local load total L		0.1w	
1039.	IPM Temperature	REC Temperature		0.1℃	No use
1040.	Battery Temperature	Battery Temperature		0.1℃	Lead acid/lithium battery temp
1041.	SP DSP Status	SP state			CHG/DisCHG
1042.	SP Bus Volt	SP BUS2 Volt		0.1V	
1043.					
发电量数据					
1044.	Etouser_today H	Energy to user today high		0.1kWh	
1045.	Etouser_today L	Energy to user today low		0.1kWh	
1046.	Etouser_total H	Energy to user total high		0.1kWh	
1047.	Etouser_total L	Energy to user total high		0.1kWh	
1048.	Etogrid_today H	Energy to grid today high		0.1kWh	
1049.	Etogrid_today L	Energy to grid today low		0.1kWh	
1050.	Etogrid_total H	Energy to grid total high		0.1kWh	
1051.	Etogrid_total L	Energy to grid total high		0.1kWh	
1052.	Edischarge1_today H	Discharge energy1 today		0.1kWh	
1053.	Edischarge1_today L	Discharge energy1 today		0.1kWh	
1054.	Edischarge1_total H	Total discharge energy1 (high)		0.1kWh	
1055.	Edischarge1_total L	Total discharge energy1 (low)		0.1kWh	
1056.	Echarge1_today H	Charge1 energy today		0.1kWh	
1057.	Echarge1_today L	Charge1 energy today		0.1kWh	
1058.	Echarge1_total H	Charge1 energy total		0.1kWh	
1059.	Echarge1_total L	Charge1 energy total		0.1kWh	
1060.	ELocalLoad_Today H	Local load energy today		0.1kWh	
1061.	ELocalLoad_Today L	Local load energy today		0.1kWh	
1062.	ELocalLoad_Total	Local load energy total		0.1kWh	

	H				
1063.	ELocalLoad_Total L	Local load energy total		0.1kWh	
1064.	dwExportLimitApparentPower	ExportLimitApparentPower H		0.1kWh	ApparentPower
1065.	dwExportLimitApparentPower	ExportLimitApparentPower L		0.1kWh	ApparentPower
1066.	/	/	/	/	reserved
Ups information (offline)					
1067.	EPS Fac	UPSfrequency	5000/6000	0.01Hz	
1068.	EPS Vac1	UPS phase R output voltage	2300	0.1V	
1069.	EPS Iac1	UPS phase R output current		0.1A	
1070.	EPS Pac1 H	UPS phase R output power (H)		0.1VA	
1071.	EPS Pac1 L	UPS phase R output power (L)		0.1VA	
1072.	EPS Vac2	UPS phase S output voltage		0.1V	
1073.	EPS Iac2	UPS phase S output current		0.1A	No use
1074.	EPS Pac2 H	UPS phase S output power (H)		0.1VA	
1075.	EPS Pac2 L	UPS phase S output power (L)		0.1VA	
1076.	EPS Vac3	UPS phase T output voltage		0.1V	
1077.	EPS Iac3	UPS phase T output current		0.1A	No use
1078.	EPS Pac3 H	UPS phase T output power (H)		0.1VA	
1079.	EPS Pac3 L	UPS phase T output power (L)		0.1VA	
1080.	Loadpercent	Load percent of UPS ouput	0-100	1%	
1081.	PF	Power factor	0-2	0.1	Primary Value+1
BMS Infomation					
1082.	BMS_StatusOld	StatusOld from BMS	Detail information, refer to document:GrowattxxSxxP ESS Protocol;		
1083.	BMS_Status	Status from BMS		W/R	
1084.	BMS_ErrorOld	Error info Old from BMS			
1085.	BMS_Error	Errorinfomation from BMS			
1086.	BMS_SOC	SOC from BMS		R SPH6K	
1087.	BMS_BatteryVoltage	Battery voltage from BMS		R SPH6K	
1088.	BMS_BatteryCurrent	Battery current from BMS			
1089.	BMS_BatteryTemperature	Battery temperature from BMS			
1090.	BMS_MaxCurr	Max. charge/discharge current from BMS (pylon)			
1091.	BMS_GaugeRM	Gauge RM from BMS			
1092.	BMS_GaugeFCC	Gauge FCC from BMS			
1093.	BMS_FW				

1094.	BMS_DeltaVolt	Delta V from BMS			
1095.	BMS_CycleCnt	Cycle Count from BMS			
1096.	BMS_SOH	SOH from BMS			
1097.	BMS_ConstantV olt	CV voltage from BMS			
1098.	BMS_WarnInfoO ld	Warning info old from BMS			
1099.	BMS_WarnInfo	Warning info from BMS			
1100.	BMS_GaugeICCu rr	Gauge IC current from BMS			
1101.	BMS_MCUVersi on	MCU Software version from BMS			
1102.	BMS_GaugeVers ion	Gauge Version from BMS			
1103.	BMS_wGaugeFR Version_L	Gauge FR Version L16 from BMS			
1104.	BMS_wGaugeFR Version_H	Gauge FR Version H16 from BMS			
1105.	BMS_BMSInfo	BMSInformation from BMS			
1106.	BMS_PackInfo	Pack Information from BMS			
1107.	BMS_UsingCap	Using Cap from BMS			
1108.	BMS_Cell1_Volt	Cell1_Voltage from BMS			
1109.	BMS_Cell2_Volt	Cell_Voltage from BMS			
...					
1123	BMS_ Cell16_Volt	Cell16_Voltage from BMS			
1124	AC Charge Energy Today H	AC Charge Energy today	kwh		Energy today
Ninth group reserved for storage power					
1125.	ACCharge Energy TodayL	AC Charge Energy today	kwh		
1126.	AC Charge Energy Total H				Energy total
1127.	ACCharge Energy Total L				
1128.	AC Charge Power H	AC Charge Power	W		
1129.	AC Charge Power L	AC Charge Power	w		
1130.	70% INV Power adjust	uwGridPower_70_AdjEE_SP	W		

1131.	Extra AC Power to grid_H	Extra inverte AC Power to grid High	For SPA connect inverter		SPA used
1132.	Extra AC Power to grid_L	Extrainverte AC Power to grid Low			SPA used
1133.	Eextra_today H	Extra inverter PowerTOUser_Extra today (high)	R	0.1kWh	SPA used
1134.	Eextra_today L	Extra inverter PowerTOUser_Extra today (low)	R	0.1kWh	SPA used
1135.	Eextra_total H	Extra inverter PowerTOUser_Extra total(high)		0.1kWh	SPA used
1136.	Eextra_total L	Extra inverter PowerTOUser_Extra total(low)		0.1kWh	SPA used
1137.	Esystem_today H	System electric energy today H		0.1kWh	SPA used System electric energy today H
1138.	Esystem_ today L	System electric energy today L		0.1kWh	SPA used System electric energy today L
1139.	Esystem_total H	System electric energy total H		0.1kWh	SPA used System electric energy total H
1140.	Esystem_total L	System electric energy total L		0.1kWh	SPA used System electric energy total L
.....	/	/	/	/	reversed
1249.	/	/	/	/	reversed

thirteen group for Storage power's SPA					
2000	Inverter Status	Inverter run state	0:waiting, 1:normal, 3:fault		SPA
.....	reversed				
2035	Pac H	Output power (high)		0.1W	SPA
2036	Pac L	Output power (low)		0.1W	SPA
2037	Fac	Grid frequency		0.01Hz	SPA
2038	Vac1	Three/single phase grid voltage		0.1V	SPA
2039	Iac1	Three/single phase grid output current		0.1A	SPA
2040	Pac1 H	Three/single phase grid output watt VA (high)		0.1VA	SPA
2041	Pac1 L	Three/single phase grid output watt		0.1VA	SPA

0	Auto test stop	0	No test
1	Auto test starting	1	Testing grid volt high pro
2	Auto testing	2	Testing grid volt low pro
		3	Testing grid frequency high pro
		4	Testing grid frequency low pro

&*4: The variable “wAutoTestResult” and “cTestStepStop”: wAutoTestResult is the step test time counter, when it reach cTestStepStop, this step test will stop and fail.

&*5: Inverter Model: A , could be show: “A1 B0 D0 T0 PF U1 M5 S1” or “1000F151”

Ax=(A&0XF0000000)>>28

Bx=(A&0XF0000000)>>24

Dx=(A&0XF000000)>>20

Tx=(A&0XF00000)>>16

Px=(A&0x00F000)>>12

Ux=(A&0x000F00)>>8

Mx=(A&0x0000F0)>>4

Sx=(A&0x00000F)

&*6: DTC(Device type code)

Code No.	Device type	Note
001xx	Inverter	1 tracker and 1phase Grid connect PV inverter TL
002xx	Inverter	2 tracker and 1phase Grid connect PV inverter TL
003xx	Inverter	1 tracker and 1phase Grid connect PV inverter HF
004xx	Inverter	2 tracker and 1phase Grid connect PV inverter HF
005xx	Inverter	1 tracker and 1phase Grid connect PV inverter LF
006xx	Inverter	2 tracker and 1phase Grid connect PV inverter LF
007xx	Inverter	1 tracker and 3phase Grid connect PV inverter TL
008xx	Inverter	2 tracker and 3phase Grid connect PV inverter TL
009xx	Inverter	1 tracker and 3phase Grid connect PV inverter LF
010xx	Inverter	2 tracker and 3phase Grid connect PV inverter LF
.....		
10001	Data logger	RF-ShineVersion
10002	Data logger	Web-ShinePano
10003	Data logger	Web-ShineWebBox
10004	Data logger	WL-WIFI Module
.....		
11001	Confluence box	Confluence box 1
.....		

&*7: Grid network power control command password:

Inverter is in lock state after power on; change the power control by network command should unlock inverter first; default pw is XXXXXX;

Unlock: send 0 to 3-135, then send password to 3-136~138; inverter will auto lock in 5min after unlocked;

Change PW: unlock first, then send 1 to 3-135, then send new password to 3-136~138;

Lock: send 0 or 2 to 3-135;

&*8: Inverter fault code and warning code

Fault code		Warning code	
0x00000001	\	0x0001	Fan warning
0x00000002	Communication error	0x0002	String communication abnormal
0x00000004	\	0x0004	StrPIDconfig Warning
0x00000008	StrReverse or StrShort fault	0x0008	\
0x00000010	Model Init fault	0x0010	DSP and COM firmware unmatched
0x00000020	Grid Volt Sample different	0x0020	\
0x00000040	ISO Sample different	0x0040	SPD abnormal
0x00000080	GFCI Sample different	0x0080	GND and N connect abnormal
0x00000100	\	0x0100	PV1 or PV2 circuit short
0x00000200	\	0x0200	PV1 or PV2 boost driver broken
0x00000400	\	0x0400	\
0x00000800	\	0x0800	\
0x00001000	AFCI Fault	0x1000	\
0x00002000	\	0x2000	\
0x00004000	AFCI Module fault	0x4000	\
0x00008000	\	0x8000	\
0x00010000	\		
0x00020000	Relay check fault		
0x00040000	\		
0x00080000	\		
0x00100000	\		
0x00200000	Communication error		
0x00400000	Bus Voltage error		
0x00800000	AutoTest fail		
0x01000000	No Utility		
0x02000000	PV Isolation Low		
0x04000000	Residual I High		
0x08000000	Output High DCI		
0x10000000	PV Voltage high		
0x20000000	AC V Outrange		
0x40000000	AC F Outrange		
0x80000000	TemperatureHigh		

&*9 Warning Value

	Warning Value 1	Warning Value 2	Warning Value 3
0x0001	String1abnormal	PV1ShortCircuit	AC SPD abnormal
0x0002	String2abnormal	PV2ShortCircuit	DC SPD abnormal
0x0004	String3abnormal	PV3ShortCircuit	
0x0008	String4abnormal	PV4ShortCircuit	
0x0010	String5abnormal	PV5ShortCircuit	
0x0020	String6abnormal	PV6ShortCircuit	
0x0040	String7abnormal	PV7ShortCircuit	
0x0080	String8abnormal	PV8ShortCircuit	
0x0100	String9abnormal	BT1DriverFault	
0x0200	String10abnormal	BT2DriverFault	
0x0400	String11abnormal	BT3DriverFault	
0x0800	String12abnormal	BT4DriverFault	
0x1000	String13abnormal	BT5DriverFault	
0x2000	String14abnormal	BT6DriverFault	
0x4000	String15abnormal	BT7DriverFault	
0x8000	String16abnormal	BT8DriverFault	

&*11: Inverter Model: A , could be show: “S0A D01 B01 T06 P0F U01 M03E8” or “0A0101060F0103E8”

```

Sx=(A&0XFF00000000000000)>>56
Dx=(A&0X00FF000000000000)>>48
Bx=(A&0X0000FF0000000000)>>40
Tx=(A&0X000000FF00000000)>>32
Px=(A&0x00000000FF000000)>>24
Ux=(A&0x0000000000FF0000)>>16
Mx=(A&0x000000000000FFFF)

```

HybridAbnoram/Fault/warning bit definition

(Abnormal:record event for debug,continueworking;fault:record event and show for debug,stopworking;Warning:record event and show,continue working)

Word definition		Bit definition		comment
System fault word0	Byte0	MasterForceINVFault	0.	M3 on/off control
		MasterForceSPFault	1.	
		BusVoltHigh_TZ	2.	restart PWM
		BusVoltHigh_ISR	3.	restartPWM
		reserved	4.	
		reserved	5.	
		reserved	6.	

	Byte1	reserved	7.	Grid side abnormal
		GridZClossFault	8.	
		reserved	9.	
		reserved	10.	
		GFCIHigh	11.	
		GridR_VFault	12.	
		GridS_VFault	13.	
		GridT_VFault	14.	
		GridFFault	15.	
System fault word1	Byte2	RelayFault	0.	Grid side abnormal
		GFCIDamage	1.	
		GridR_VLowFault	2.	
		GridR_VHighFault	3.	
		GridS_VLowFault	4.	
		GridS_VHighFault	5.	
		GridT_VLowFault	6.	
		GridT_VHighFault	7.	
	Byte3	INVCurrOCP_ISR	8.	Grid side abnormal
		INVCurrOCP_TZ	9.	
		DCIHigh	10.	
		reserved	11.	
		INVR_CurrOCP_Rms	12.	
		INVS_CurrOCP_Rms	13.	
		INVT_CurrOCP_Rms	14.	
		NoUtility	15.	
System fault word2	Byte4	GridFlowFault	0.	Grid side abnormal
		GridFHighFault	1.	
		GridVolt_Unbalance_Fault	2.	
		AC_PLL_Fault	3.	
		OverLoadFault	4.	
		reserved	5.	
		reserved	6.	
		reserved	7.	
	Byte5	EPS_LineVoltR_Loss	8.	EPS side abnormal
		EPS_LineVoltS_Loss	9.	
		EPS_LineVoltT_Loss	10.	
		reserved	11.	
		reserved	12.	
		reserved	13.	
		reserved	14.	
		reserved	15.	

System fault word3	Byte6	BatTerminalReversed	0.	BAT Side abnormal
		BatTerminalOpen	1.	
		BMS Battery Open		
		BatteryVoltageLow	2.	
		BatteryVoltageHigh	3.	
		reserved	4.	
		reserved	5.	
		reserved	6.	
		reserved	7.	
	Byte7	reserved	8.	BAT Side abnormal
		reserved	9.	
		reserved	10.	
		reserved	11.	
		reserved	12.	
		reserved	13.	
		reserved	14.	
		reserved	15.	
System fault word4	Byte8	reserved	0.	PV Side Abnormal
		reserved	1.	
		reserved	2.	
		reserved	3.	
		reserved	4.	
		PV1_VoltLowWarn	5.	
		PV2-VoltLowWarn	6.	
		reserved	7.	
	Byte9		8.	PV Side Abnormal
			9.	
			10.	
		reserved	11.	
		reserved	12.	
		reserved	13.	
		reserved	14.	
		reserved	15.	
System fault word5	Byte10	NEDetectFault	0.	Sytem fault
		PVISOFault	1.	
		reserved	2.	
		BusVoltHighFault_ISR	3.	
		BusSampleFault	4.	
		UHCTFault	5.	
		AComFault	6.	
		BComFault	7.	

	Byte11	BusVoltHighFault_TZ	8.	Sytem fault
		AuotTestFault	9.	
		DCHigh	10.	
		NTCOpenFault	11.	
		reserved	12.	
		BBHeatsink_TempOver	13.	
		BBOCP_FaultISR	14.	
		BBOCP_FaultTZ	15.	
System fault word6	Byte12	PV1_VoltHighFault	0.	Sytem fault
		PV2_VoltHighFault	1.	
		BHeatsink_Overtemp	2.	
		INVHeatsink_Overtemp	3.	
		reserved	4.	
		reserved	5.	
		reserved	6.	
		reserved	7.	
	Byte13	BoostDriver1Warn	8.	System warning
		BoostDriver2Warn	9.	
		WARN104	10.	
		PV1_ShortFault	11.	
		PV2_ShortFault	12.	
		Meter Comm Loss	13.	
		PairingTimeOut	14.	
		CT LN Reversed	15.	
System fault word6	Byte14	BMS COM Fault	0.	
		BMS Error: xxx	1.	
		Battery reversed	2.	
		BAT NTC Open	3.	
		SS Timeout	4.	
		Bat voltage low	5.	
		Bat T Outrange	6.	
		BATOutput_Overload	7.	
	Byte15	reserved	8.	
		reserved	9.	
		reserved	10.	
		reserved	11.	
		reserved	12.	
		reserved	13.	
		reserved	14.	
		reserved	15.	
System fault word7		reserved		

5 Set address

Refer to the Inverter user manual. Always is :

Knock the pv inverter to let the lcd display to the “COM Addr: xxx”, then double knock, if displays “Move”, you should another double knock, until it displays a address number, then you can give a single knock to change the address, this address will be remembered when the lcd backlight off.

6 Notice

- 1) It can drive mostly 32 pv inverters for one rs485 comport.
- 2) There are only read input and hold registers commands even the newest version.
- 3) App user could only care the input register.
- 4) App user could not care the holding registers.
- 5) Except the CEI0-21 and VDE-AR-N 4105 power management registers, you should refer the manufactory’s suggestion when writing the other registers;